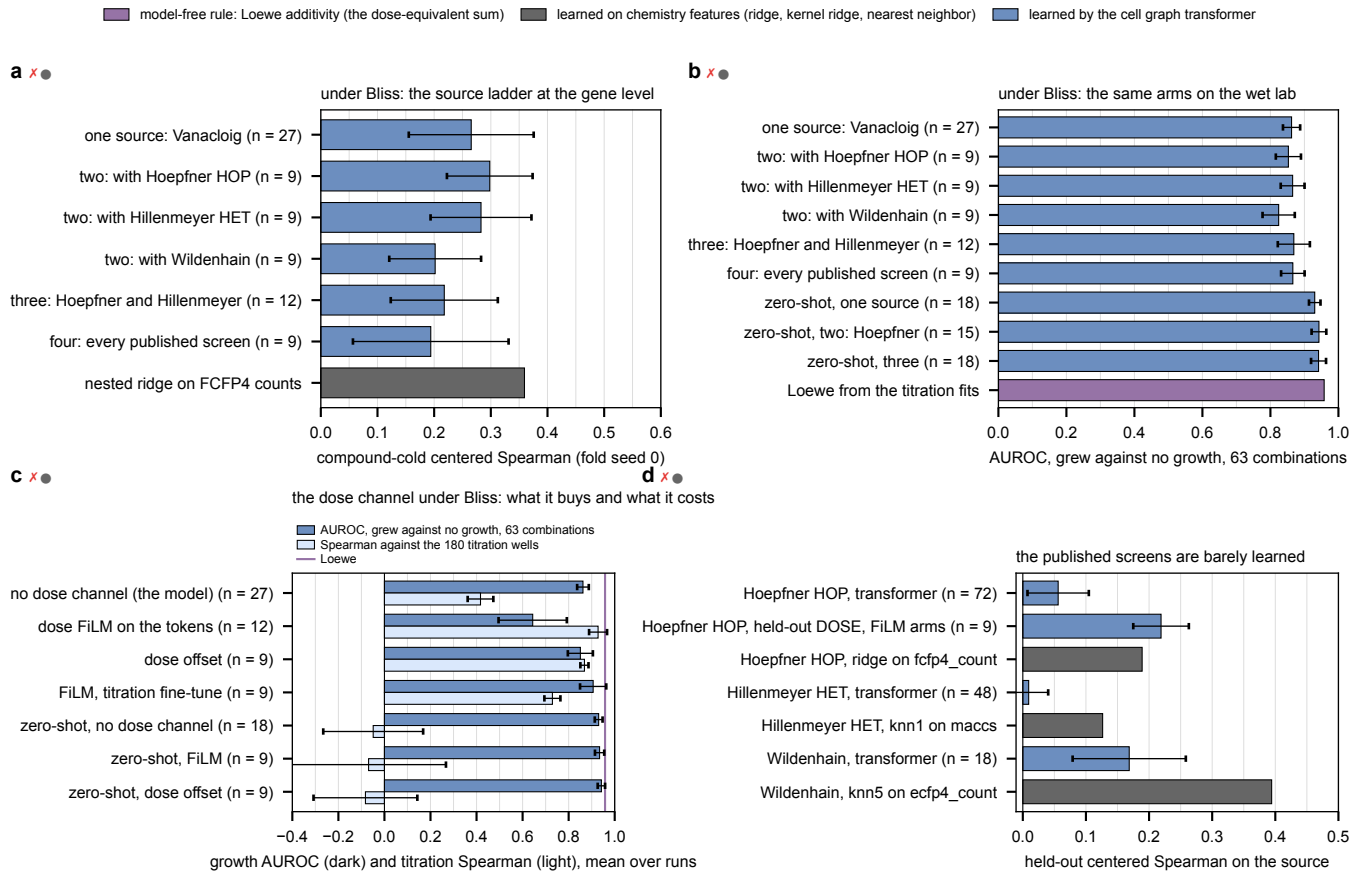


Figure 10: Supplement eight: round ten, the Bliss cell readout at three seeds. Every arm is three seeds on the three screening folds of fold seed 0, read on both machines where it ran on both; the bar is the mean over runs, the whisker the standard deviation. **a**, The source ladder on the published side: one source 0.265, with Hoepfner 0.298, with Hillenmeyer 0.283, with Wildenhain 0.202, three 0.218, four 0.194, run spreads 0.110 to 0.137, against nested ridge at 0.359. No published screen lifts the target, and the order is more sources, lower, inside one spread. ✗● **b**, The same arms on the wet lab, growth AUROC over the 63 combinations: the fitted ladder reads 0.862 to 0.866 and does not move with sources; the zero-shot ladder (anchors only, never a titration well) reads 0.930, 0.943 and 0.942 on one, two and three sources, every value above the Bliss rule on the measured singles (0.872) and the last at Loewe's (0.958). Three runs per arm; the steps are inside one spread, so this is a measured ordering rather than a cleared difference. ✗● **c**, The dose channel: each arm's growth AUROC (dark) beside the Spearman of its host prediction against the 180 titration wells (light), a fit for the arms trained on the titration and a prediction for the zero-shot arms. FiLM on the tokens reads the titration at 0.928 and composes at 0.644; the dose offset reads it at 0.869 and composes at 0.851, against 0.862 with no channel; the titration fine-tune with FiLM reads 0.729 and composes at 0.906. Zero-shot, no channel reads the titration, and the dose offset composes at 0.942. ✗● **d**, What the Bliss arms predict on the published screens they learn from, the held-out compound score per source over every round-ten arm that carries it: Hoepfner 0.056 against 0.189, Hillenmeyer 0.009 against 0.126, Wildenhain 0.169 against 0.394; the held-out Hoepfner dose under FiLM, 0.219. The screens are not learned at a held-out compound, so a screen could not have helped the target at one. ✗●